

使用四变量映射简化以下布尔表达式

3.6 Simplify the following Boolean expressions, using four-variable maps:

(a)* $A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$

(b)* $x'z + w'xy' + w(x'y + xy')$

(c) $A'B'C'D + AB'D + A'BC' + ABCD + AB'C$

(d) $A'B'C'D' + BC'D + A'C'D + A'BCD + ACD'$

找出下列布尔函数的所有原蕴含项，并确定哪些是本质项

3.9 Find all the prime implicants for the following Boolean functions, and determine which are essential:

(a)* $F(w, x, y, z) = \Sigma(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$

(b)* $F(A, B, C, D) = \Sigma(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$

(c) $F(A, B, C, D) = \Sigma(2, 3, 4, 5, 6, 7, 9, 11, 12, 13)$

(d) $F(w, x, y, z) = \Sigma(1, 3, 6, 7, 8, 9, 12, 13, 14, 15)$

(e) $F(A, B, C, D) = \Sigma(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

(f) $F(w, x, y, z) = \Sigma(0, 1, 2, 5, 7, 8, 10, 15)$

将以下布尔函数从乘积和形式转换为简化后的乘积形式。

3.11 Convert the following Boolean function from a sum-of-products form to a simplified product-of-sums form.

$$F(x, y, z) = \Sigma(0, 1, 2, 5, 8, 10, 13)$$

简化下列布尔函数：

3.12 Simplify the following Boolean functions:

(a)* $F(A, B, C, D) = \Pi(1, 3, 5, 7, 13, 15)$

(b) $F(A, B, C, D) = \Pi(1, 3, 6, 9, 11, 12, 14)$

将下列函数简化，并用两级 NAND（与非门）门电路实现

3.16 Simplify the following functions, and implement them with two-level NAND gate circuits:

(a) $F(A, B, C, D) = AC'D' + A'C + ABC + AB'C + A'C'D'$

(b) $F(A, B, C, D) = A'B'C'D + CD + AC'D$

(c) $F(A, B, C) = (A' + C' + D')(A' + C')(C' + D')$

(d) $F(A, B, C, D) = A' + B + D' + B'C$

将下列函数简化，并用两级 NOR（或非门）门电路实现：

3.19 Simplify the following functions, and implement them with two-level NOR gate circuits:

(a)* $F = wx' + y'z' + w'yz'$

(b) $F(w, x, y, z) = \Sigma(0, 3, 12, 15)$

(c) $F(x, y, z) = [(x + y)(x = z)]'$

绘制以下表达式的多级 NAND（与非门）电路

3.21 Draw the multiple-level NAND circuit for the following expression:
 $w(x + y + z) + xyz$

将图 4.4 所示电路的逻辑图转换为多级 NAND 电路

3.22 Convert the logic diagram of the circuit shown in Fig. 4.4 into a multiple-level NAND circuit.

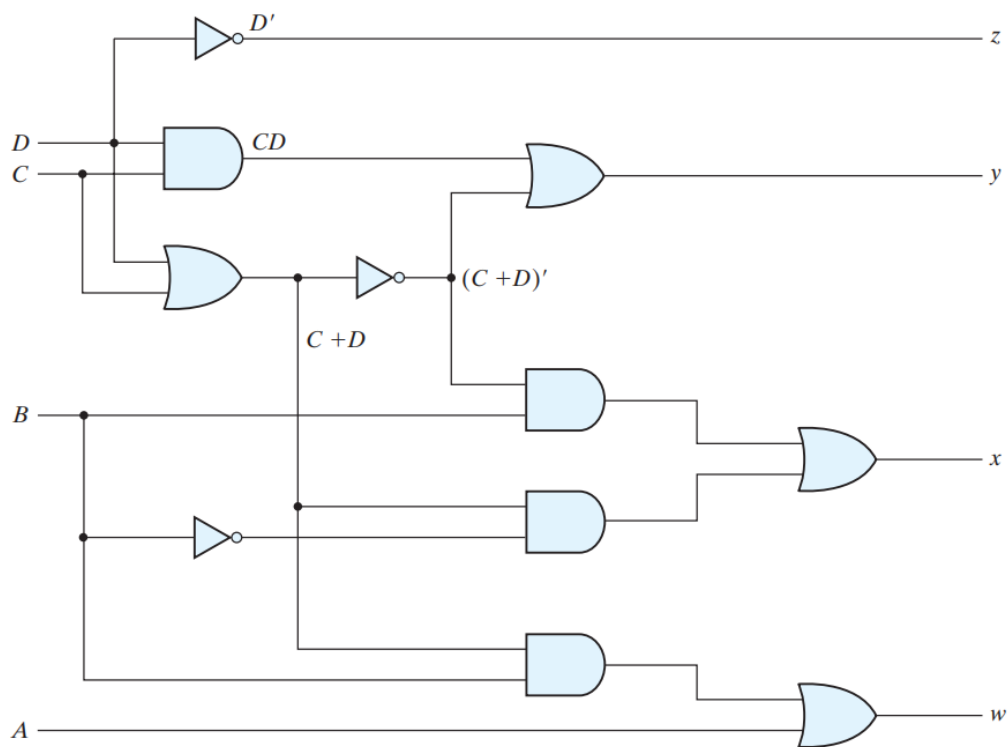
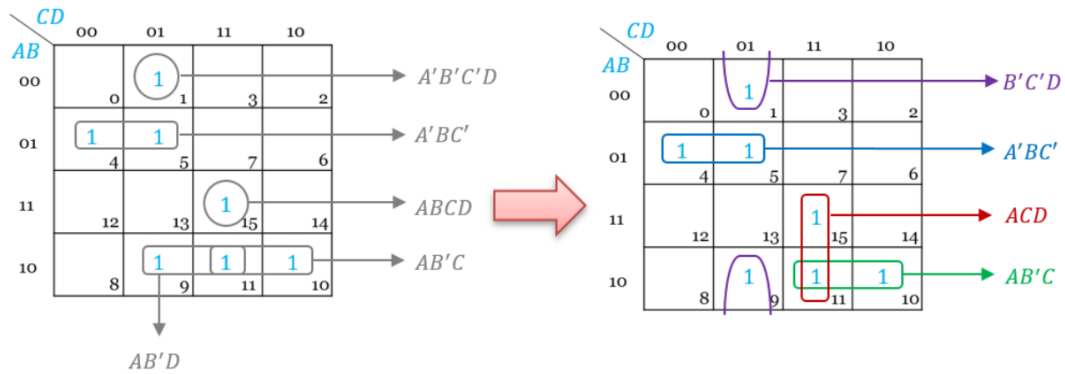


FIGURE 4.4

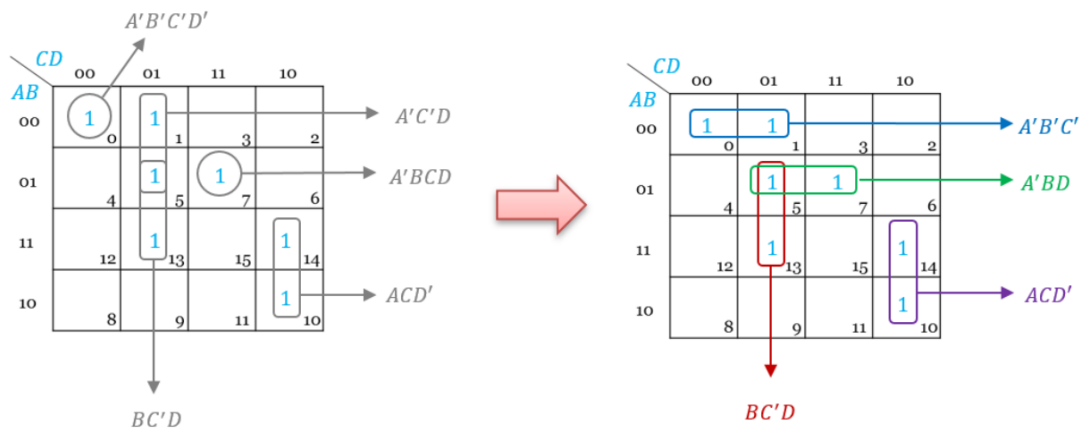
3.6C

$$F(A, B, C, D) = A'BC' + B'C'D + ACD + AB'C$$



3.6D

$$F(A, B, C, D) = A'B'C' + A'BD + BC'D + ACD'$$



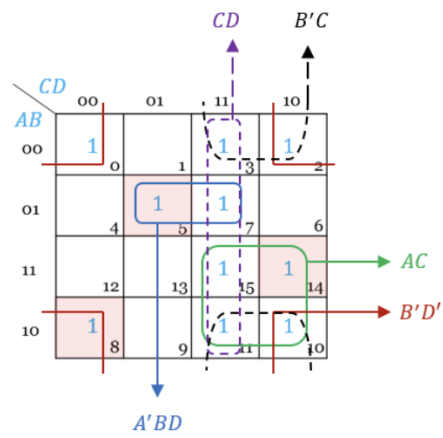
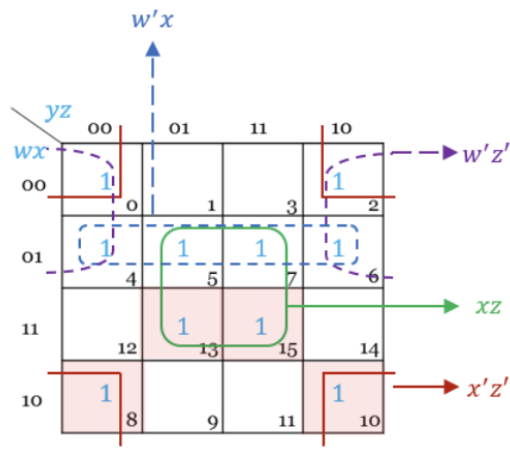
(c) $F(A, B, C, D) = A'BC' + B'C'D + ACD + AB'C$.

(d) $F(A, B, C, D) = A'B'C' + A'BD + BC'D + ACD'$.

3.9a-f

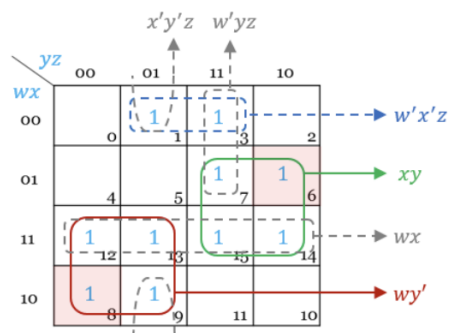
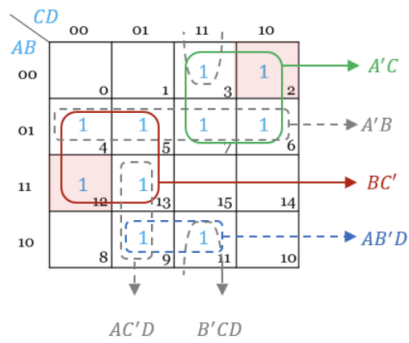
Aaaaaaa

bbbbbbb



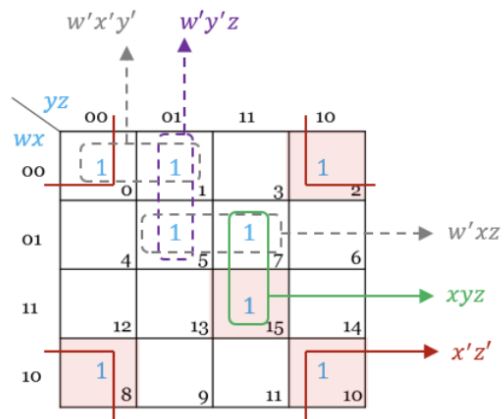
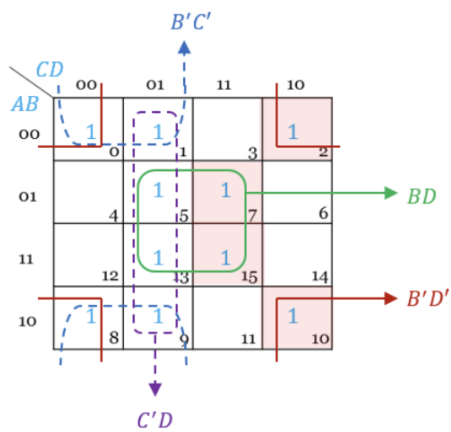
Cccccccccccccccc

DDDDDDDDDDDDDDDDDDDD



eeeeeeeeeeeeee

ffffffffffffffff



(a)

Prime implicants $\Rightarrow xz, x'z', w'x$ and $w'z'$

Essential prime implicants $\Rightarrow xz$ and $x'z'$

(b)

Prime implicants $\Rightarrow AC, B'D', A'BD, CD$ and $B'C$

Essential prime implicants $\Rightarrow AC, B'D'$ and $A'BD$

(c)

Prime implicants $\Rightarrow A'C, BC', AB'D, AC'D, B'CD$ and $A'B$

Essential prime implicants $\Rightarrow A'C$ and BC'

(d)

Prime implicants $\Rightarrow xy, wy', w'x'z, wx, x'y'z$ and $w'yz$

Essential prime implicants $\Rightarrow xy$ and wy'

(e)

Prime implicants $\Rightarrow BD, B'D', C'D$ and $B'C'$

Essential prime implicants $\Rightarrow BD$ and $B'D'$

(f)

Prime implicants $\Rightarrow x'z', xyz, w'y'z, w'xz$ and $w'x'y'$

Essential prime implicants $\Rightarrow x'z'$ and xyz

3.11

The 1's of the given function:

$$F = \sum(0, 1, 2, 5, 8, 10, 13)$$

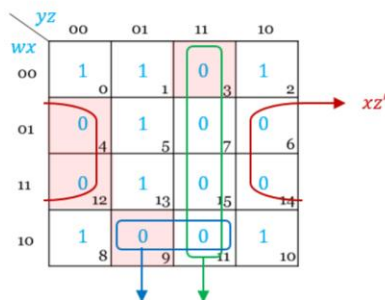
are plotted as shown below and we fill in the remaining squares with 0's.

Because the 0's of F are 1's of F' , the minimum sum of products for F' can be determined by looping the 0's on the map of F . Then, we shade all 0's that can be covered by only one prime implicant which are 0₃, 0₄, 0₉ and 0₁₂ as shown below. Then,

$$F' = xz' + yz + wx'z$$

The complement of the minimum sum of products for F' is the minimum product of sums for F . Thus, the minimum product of sums for F is

$$\begin{aligned} F &= (F')' \\ &= (xz' + yz + wx'z)' \quad (\text{DeMorgan}) \\ &= (x' + z)(y' + z')(w' + x + z') \end{aligned}$$



3.12

$$F(A, B, C, D) = \prod(1, 3, 5, 7, 13, 15)$$

		CD			
		00	01	11	10
AB	00	1	0	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	1	1	1

$$F = D' + AB'$$

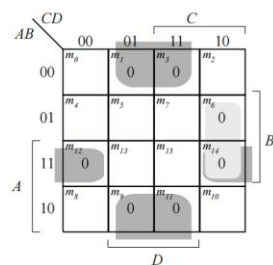
(b)

$$F = \prod(1, 3, 6, 9, 11, 12, 14)$$

$$F' = \sum(1, 3, 6, 9, 11, 12, 14)$$

$$F = \sum(0, 2, 4, 5, 7, 8, 10, 13, 15)$$

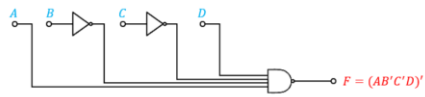
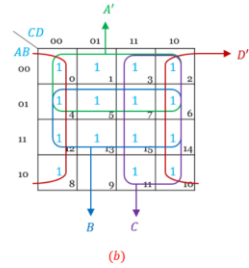
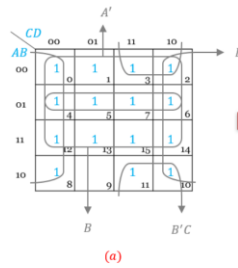
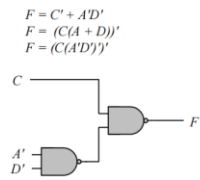
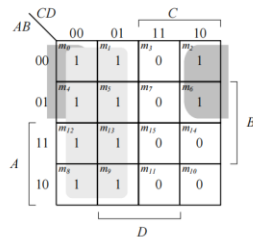
$$F = BD + B'D' + A'BC' \text{ (or } A'CD')$$



3.16 C

D

c) $F = (A' + C' + D')(A' + C')(C' + D')$
 $F' = (A' + C' + D')' + (A' + C')' + (C' + D')'$
 $F' = ACD + AC + CD$



3.19BC

(b)

First, the 1's of F :

$$F = \sum(0, 3, 12, 15)$$

are plotted as shown below in (a). Then, from the 0's,

$$F' = y'z + yz' + w'x + wx'$$

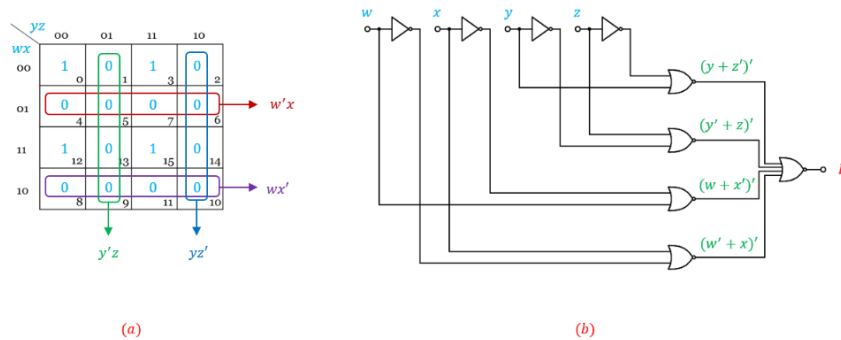
Therefore, the simplified function in the minimum-product-of-sums form is

$$F = (y + z')(y' + z)(w + x')(w' + x)$$

To implement the function F using two-level NOR gate circuit, we reformulate the function as follows:

$$\begin{aligned} F &= (F')' \\ &= \left\{ [(y + z')(y' + z)(w + x')(w' + x)]' \right\}' \\ &= \{ (y + z')' + (y' + z)' + (w + x')' + (w' + x)' \}' \end{aligned}$$

Based on the last equation, we implement the function F using two-level NOR gate circuit as shown below in (b).



(c)

There is an error in the given function. We assume the correction to be

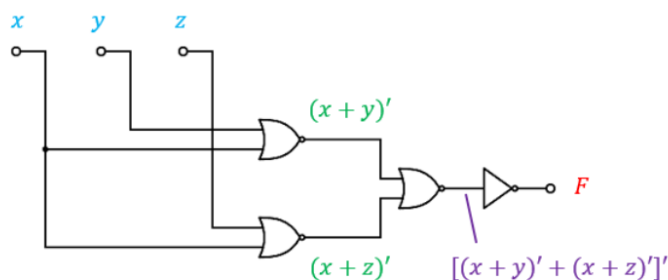
$$F(x, y, z) = [(x + y)(x + z)]'$$

Next, we formulate this function so that it can be implemented using two-level NOR gate circuit:

$$\begin{aligned} F(x, y, z) &= [(x + y)(x + z)]' \\ &= (x + y)' + (x + z)' \end{aligned}$$

But $F = (F')'$; hence,

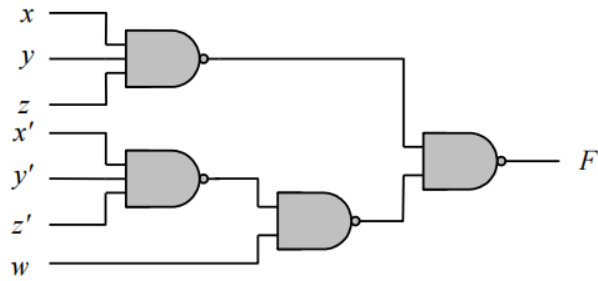
$$F(x, y, z) = \left\{ [(x + y)' + (x + z)']' \right\}'$$



3.21

$$F = w(x + y + z) + xyz$$

$$F' = [w(x + y + z)]'[xyz]' = [w(x'y'z')']'(xyz)'$$



$$\begin{aligned}
 z &= D' \\
 y &= CD + (C + D)' \\
 x &= B(C + D)' + B'(C + D) \\
 w &= A + B(C + D)
 \end{aligned}$$

步骤2

步骤2/4

We formulate these functions so that it can be implemented using NAND gates as follows:

First, the function $(C + D)$ can be expressed as

$$\begin{aligned}
 C + D &= [(C + D)']' \\
 &= [C'D']' \quad (1)
 \end{aligned}$$

Next, for y ,

$$\begin{aligned}
 y &= (y')' \\
 &= \{ [CD + (C + D)']' \}' \\
 &= \{ [CD + (C'D)']' \}' \Rightarrow \text{(From Eq.1)} \\
 &= \{ (CD)' \cdot (C'D)'\}'
 \end{aligned}$$

For x ,

$$\begin{aligned}
 x &= (x')' \\
 &= \{ [B(C + D)' + B'(C + D)]' \}' \Rightarrow \text{(From Eq.1)} \\
 &= \{ [B(C'D) + B'(C'D)']' \}' \\
 &= \{ [BC'D]' \cdot [B'(C'D)']' \}'
 \end{aligned}$$

For w ,

$$\begin{aligned}
 w &= (w')' \\
 &= \{ [A + B(C + D)]' \}' \Rightarrow \text{(From Eq.1)} \\
 &= \{ [A + B(C'D)']' \}' \\
 &= \{ A' \cdot [B(C'D)']' \}'
 \end{aligned}$$

We implement the expressions for z , y , x , and w using the multiple-level NAND circuit as shown below.

